

REGRESSION ANALYSIS (LECTURE 5)



DR. KOUSHIK GHOSH

Department of Mathematics, University Institute of
Technology, The University of Burdwan,
Golapbag (North), Burdwan-713104, India

E-mail:

koushikg123@yahoo.co.uk/koushikg123@gmail.com

Simple Nonlinear (Quadratic) Regression Using Method of Least Squares:

We assume the equation of Quadratic (Parabolic) Regression Curve of Y on X as $y = a + bx + cx^2$ and the equation of Quadratic (Parabolic) Regression Curve of X on Y as $x = d + ey + fy^2$. Let us assume that we are given with a bivariate data (x_i, y_i) ; $i = 1, 2, 3, \dots, N$ for the variables X and Y . We have to estimate the parameters a, b, c, d, e and f using the given bivariate data applying the Method of Least Squares. Applying the Method of Least Squares for the Quadratic (Parabolic) Regression Curve of Y on X we get the normal equations as:

$$\begin{aligned}\sum_{i=1}^N y_i &= Na + b \sum_{i=1}^N x_i + c \sum_{i=1}^N x_i^2 \\ \sum_{i=1}^N x_i y_i &= a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2 + c \sum_{i=1}^N x_i^3 \\ \sum_{i=1}^N x_i^2 y_i &= a \sum_{i=1}^N x_i^2 + b \sum_{i=1}^N x_i^3 + c \sum_{i=1}^N x_i^4\end{aligned}$$

Solving the above system we can get the corresponding values of a, b and c .

Again applying the Method of Least Squares for the Quadratic (Parabolic) Regression Curve of X on Y we get the normal equations as:

$$\sum_{i=1}^N x_i = Nd + e \sum_{i=1}^N y_i + f \sum_{i=1}^N y_i^2$$

$$\sum_{i=1}^N x_i y_i = d \sum_{i=1}^N y_i + e \sum_{i=1}^N y_i^2 + f \sum_{i=1}^N y_i^3$$

$$\sum_{i=1}^N x_i y_i^2 = d \sum_{i=1}^N y_i^2 + e \sum_{i=1}^N y_i^3 + f \sum_{i=1}^N y_i^4$$

Solving the above system we can get the corresponding values of d , e and f .

Problem: Obtain both the quadratic (parabolic) regression equations for the given bivariate data using Method of Least Squares:

X : 3 2 7 5 8

Y : 6 1 8 6 9

Also estimate y for $x = 11$ and x for $y = 10$.

Answer: Here we have to find out the two quadratic (parabolic) regression equations Y on X and X on Y . To estimate y for $x = 11$ we will use parabolic regression equation of Y on X and to estimate x for $y = 10$ we will use parabolic regression equation of X on Y .

Answer:

Given bivariate data:

X	3	2	7	5	8
Y	6	1	8	6	9

We produce the following table:

x	y	x ²	x ³	x ⁴	y ²	y ³	y ⁴	xy	x ² y	xy ²	
3	6	9	27	81	36	216	1296	18	54	108	
2	1	4	8	16	1	1	1	2	4	2	
7	8	49	343	2401	64	512	4096	56	392	448	
5	6	25	125	625	36	216	1296	30	150	180	
8	9	64	512	4096	81	729	6561	72	576	648	
Σ	25	30	151	1015	7219	218	1674	13250	178	1176	1386

for the parabolic regression equation

for the parabolic regression equation of Y on X we have the normal equations as

$$\Sigma y = Na + b \Sigma x + c \Sigma x^2$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3$$

$$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4$$

$$\text{or, } 30 = 5a + 25b + 151c$$

$$178 = 25a + 151b + 1015c$$

$$1176 = 151a + 1015b + 7219c$$

Solving we get $a = -2.752$, $b = 2.7776$, $c = -0.1701$

For the parabolic regression equation of X on Y we have the normal equations as

$$\Sigma X = Nd + e \Sigma y + f \Sigma y^2$$

$$\Sigma xy = d \Sigma y + e \Sigma y^2 + f \Sigma y^3$$

$$\Sigma xy^2 = d \Sigma y^2 + e \Sigma y^3 + f \Sigma y^4$$

$$\text{or, } 25 = 5d + 30e + 218f$$

$$178 = 30d + 218e + 1674f$$

$$1386 = 218d + 1674e + 13250f$$

Solving we get $d = 2.2759$, $e = -0.415$, $f = 0.1196$

So we get the equation of the regression curve of Y on X as

$$Y = -2.752 + 2.7776X - 0.1701X^2$$

and the equation of the regression curve of X on Y as

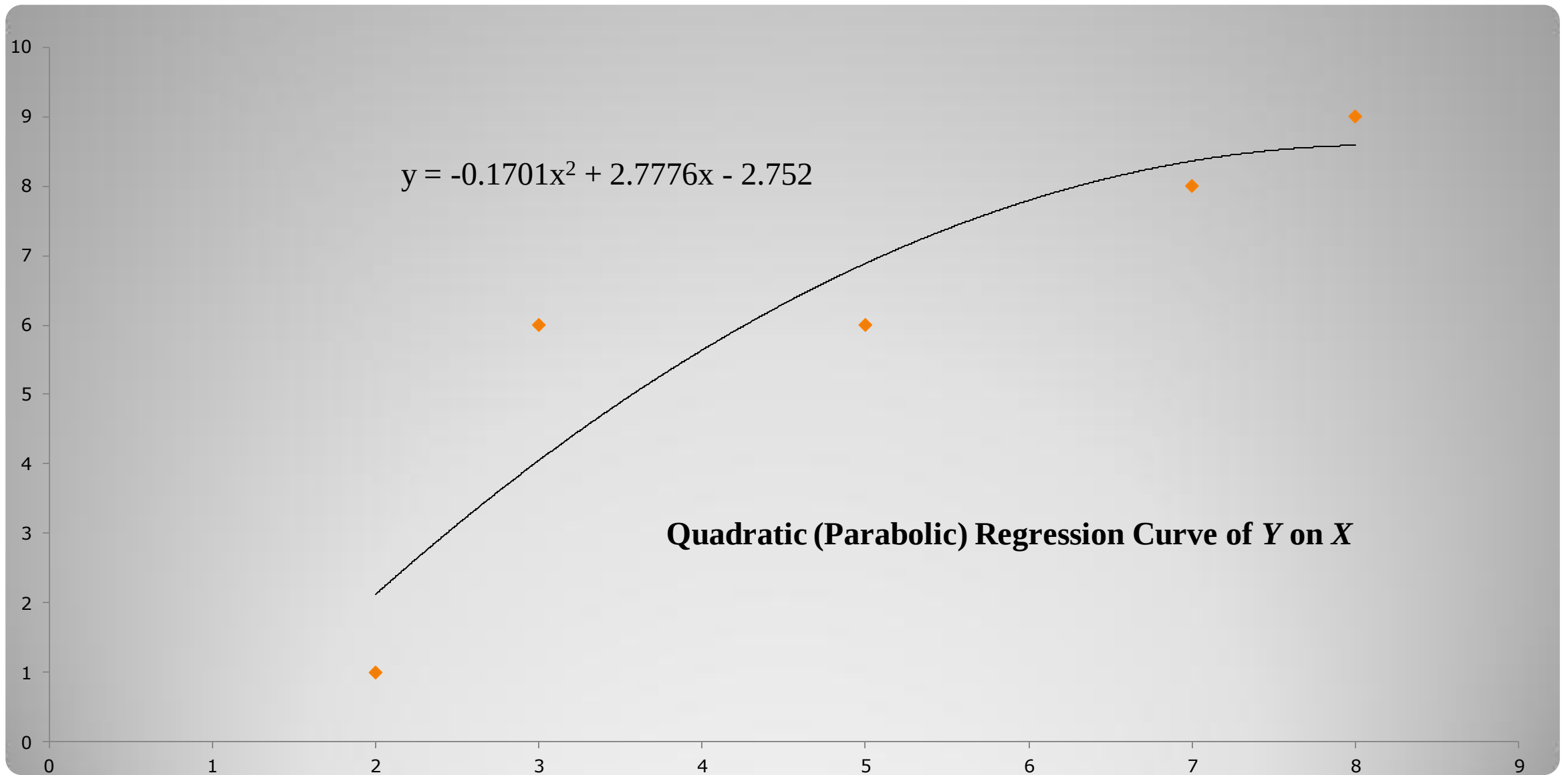
$$X = 2.2759 - 0.415y + 0.1196y^2$$

For $x=11$, we have

$$\begin{aligned} Y &= -2.752 + 2.7776 \times 11 - 0.1701 \times 11^2 \\ &= 7.2195. \end{aligned}$$

Again for $y=10$, we have

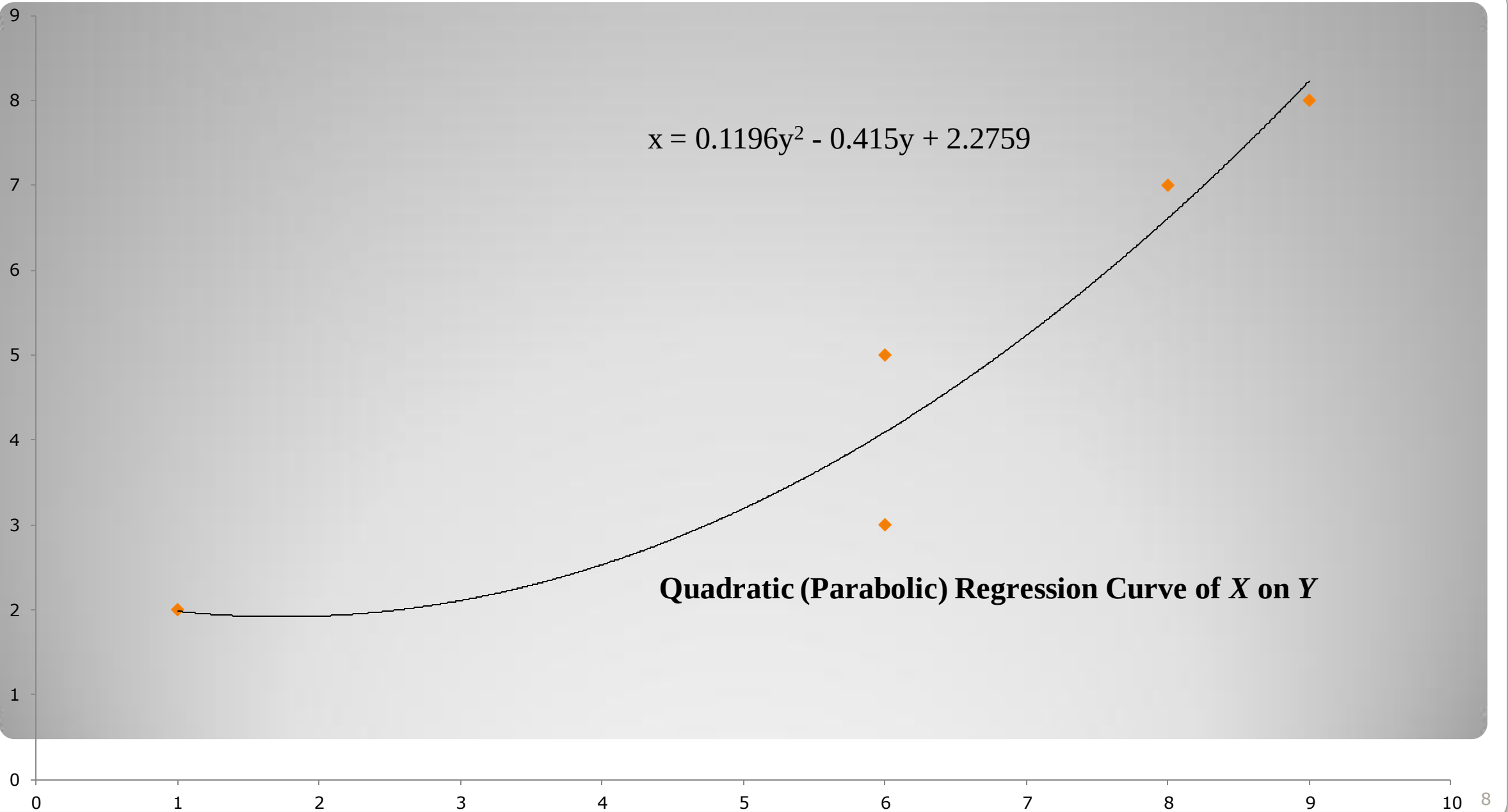
$$\begin{aligned} X &= 2.2759 - 0.415 \times 10 + 0.1196 \times 10^2 \\ &= 10.0859 \end{aligned}$$



Quadratic (Parabolic) Regression Curve of Y on X

$$x = 0.1196y^2 - 0.415y + 2.2759$$

Quadratic (Parabolic) Regression Curve of X on Y



Derivation of Normal Equations Using Least Square Method for Quadratic (Parabolic) Regression Curve of Y on X :

We consider the equation of the parabolic (quadratic) regression curve of Y on X as $y = a + bx + cx^2$. We consider the given bivariate data as $(x_i, y_i) ; i=1, 2, 3, \dots, N$. For $x=x_i$ we consider the estimated y using the parabolic regression curve as $y_i^* = a + bx_i + cx_i^2$; $i=1, 2, \dots, N$. Now the deviation of the estimated y and actually will be $y_i^* - y_i ; i=1, 2, \dots, N$.

So average (mean) squared deviation

$$S = \frac{1}{N} \sum_{i=1}^N (y_i^* - y_i)^2 = \frac{1}{N} \sum_{i=1}^N (a + bx_i + cx_i^2 - y_i)^2$$

We want to make the mean squared deviation S minimum. For that we must have

$$\frac{\partial S}{\partial a} = 0 \Rightarrow \frac{2}{N} \sum_{i=1}^N (a + bx_i + cx_i^2 - y_i) = 0$$

$$\begin{aligned} \text{or, } \sum_{i=1}^N (a + bx_i + cx_i^2 - y_i) &= 0 \\ \Rightarrow Na + b \sum_{i=1}^N x_i + c \sum_{i=1}^N x_i^2 - \sum_{i=1}^N y_i &= 0 \\ \Rightarrow \boxed{\sum_{i=1}^N y_i = Na + b \sum_{i=1}^N x_i + c \sum_{i=1}^N x_i^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial S}{\partial b} = 0 &\Rightarrow \frac{2}{N} \sum_{i=1}^N x_i (a + bx_i + cx_i^2 - y_i) = 0 \\ \Rightarrow \sum_{i=1}^N (ax_i + bx_i^2 + cx_i^3 - x_i y_i) &= 0 \Rightarrow \boxed{\sum_{i=1}^N x_i y_i = a \sum_{i=1}^N x_i + b \sum_{i=1}^N x_i^2 + c \sum_{i=1}^N x_i^3} \end{aligned}$$

$$\frac{\partial S}{\partial c} = 0 \Rightarrow \frac{2}{N} \sum_{i=1}^N x_i^2 (a + bx_i + cx_i^2 - y_i) = 0 \Rightarrow \boxed{\sum_{i=1}^N x_i^2 y_i = a \sum_{i=1}^N x_i^2 + b \sum_{i=1}^N x_i^3 + c \sum_{i=1}^N x_i^4}$$